products_03-prayog

Prayog STEM lab kit: manufacturing manual

> **Status: firmware and analysis tools tested on the host; the hub board is order-ready (all automatic checks pass); no hardware built; sensors uncalibrated; syllabus mapping and school procurement route unverified.**

1. What is made

A hub board (ATmega328P at 16 MHz, CH340C USB-serial with auto-reset, USB-C, eight 3-pin sensor sockets A0-A7, photogate port on D2, ultrasonic port on D4/D5, reaction pad button D8 + LED D7) plus **12 plug-in sensor modules** and 20 experiments with worksheets and a teacher guide. Firmware streams CSV at 115200 baud; Python tools analyse data offline.

Deliverable	Where
Hub PCB	`utility10/pcb/release/prayog/`
Firmware	`products/03-prayog/release/prayog.hex` (9.6 KB flash, 434 B RAM), source `products/03-prayog/firmware/`, 40 sensor-library checks
Analysis tools	`products/03-prayog/tools/prayog_tools.py` (11 of 11 pass on simulated data)
Experiments	`products/03-prayog/docs/experiments.md` (20)
Cost model	`products/03-prayog/docs/kit_cost.txt` (BOM Rs 2,580 at about 1,000 kits)

2. Hub board bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J1	USB-C data + 5V	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
F1	PTC 500mA	Fuse_1206_3216Metric	Littelfuse 1206L050YR (6 V)	C163512	SMT (JLC)
D3	SMAJ5.0A	D_SMA	Littelfuse SMAJ5.0A	C83329	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R1	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R2	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
C5	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
U2	CH340C	SOIC-16_3.9x9.9mm_P1.27mm	CH340C	C84681	SMT (JLC)
C6	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C7	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C8	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
Y1	16 MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	YXC X322516MOB4SI 16 MHz 12 pF 3225 (LCSC listing seen in search results; matches the 22 pF load caps)	C12668	SMT (JLC)
C9	22p	C_0603_1608Metric	Samsung CL10C220JB8NNNC C0G	C1653	SMT (JLC)
C10	22p	C_0603_1608Metric	Samsung CL10C220JB8NNNC C0G	C1653	SMT (JLC)
J3	Analog socket A0	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J4	Analog socket A1	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J5	Analog socket A2	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J6	Analog socket A3	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J7	Analog socket A4	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J8	Analog socket A5	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J9	Analog socket A6	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J10	Analog socket A7	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J11	Photogate	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J12	Ultrasonic HC-SR04	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
SW2	Reaction pad	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
D1	Green	LED_0603_1608Metric	Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check)	C72043	SMT (JLC)
R5	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)

3. The 12 modules (bought-in or built in-house)

Temperature (10k NTC with 10k divider), light (LDR with divider), sound level (electret microphone module), magnetic field (linear Hall sensor), soil moisture (capacitive), voltage probe x2 (resistor divider with input protection, low voltage only), IR reflective, water conductivity (AC-driven pair), ultrasonic distance (HC-SR04 class), photogate x2 (U-slot optical sensors, output LOW when the beam is interrupted), button + LED reaction pad. Each module plugs into a 3-pin JST-PH socket: +5 V, signal, GND.

Procure from a module supplier with lot test data, or build them on small boards using the same procedure (STANDARD_PROCEDURES A-F).

4. Make the hub

1. Order PCBs and SMT assembly per STANDARD_PROCEDURES A and B (the CH340C, ATmega328P, crystal and 22 pF caps are on the JLC BOM when released); solder the hand/THT parts (USB-C tabs, JST sockets, ISP header, button).
2. **Bootloader/firmware:** fuses for a 16 MHz external crystal: `avrdude -p m328p -c usbasp -U lfuse:w:0xFF:m -U hfuse:w:0xDE:m -U efuse:w:0xFD:m` (Arduino Uno values: full-swing 16 MHz crystal, 2.7 V brown-out, reset vector at the Optiboot bootloader; use hfuse 0xD9 if you flash the hex directly with no bootloader; confirm all values against the ATmega328P datasheet). Burn an Arduino-Uno-class (Optiboot) bootloader over ISP, then upload `prayog.hex` over USB at 115200 baud (`avrdude -p m328p -c arduino -P <port> -b 115200 -U flash:w:prayog.hex:i`), or flash the hex directly over ISP without a bootloader (then the auto-reset capacitor is unused).
3. Plug the USB-C cable: +5 V = 5.0 V, the CH340C enumerates as a serial port (install the WCH driver on Windows if needed). Open a serial terminal at 115200: CSV lines appear 10 times a second (`ms,A0..A7,echo_cm,gate_count`).
4. Commands: `SET <0-7> <NTC|LDR|MIC|HALL|SOIL|VOLT|IR|COND>` assigns a sensor type to a socket.

5. Calibration (per module type, before shipping a kit)

NTC: check 0 and 100 C reference (ice and boiling water: within the 0.6 C limit set by the 10-bit ADC); LDR: record a lux table with a reference lux meter; Hall: sensitivity with a known magnet; soil: dry/wet counts (typical 520/260: measure); ultrasonic: speed of sound correction with temperature (library does it) checked at 100 cm; photogate: timing check with a falling-object rig. **Sensor constants in the library are typical datasheet values: the experiments must be validated with real sensors (the analysis validation used simulated data only).**

6. End-of-line test per hub (and per kit)

1. Supply current from USB under 100 mA with no modules.
2. Serial output appears; each socket reads a pot or fixed divider correctly (A0-A7 within +-1 %).
3. Photogate: interrupt the beam: a `G,<us>,1` line is printed, release: `G,<us>,0`.
4. Ultrasonic: 100 cm target reads 100 +-2 cm at 20 C.
5. Reaction pad: button and LED respond.
6. Run one analysis (`prayog_tools.py`) on a recorded file. Record on `manuals/templates/eol_test_sheet.md`.

7. Kit assembly and packaging

Tray with the hub, 12 modules, cables (3-pin leads), magnets, steel ball and string for the pendulum/photogate experiments, printed student worksheets (20) and a teacher guide (to be written from `experiments.md`). Label

12+, low-voltage only, small-parts warning. Compliance to verify: BIS/lab view on whether kits for under-14s fall under the toys QCO, e-waste EPR.

8. Procurement and pilot (not done)

Ask five schools and one lab-kit dealer how kits are bought (GeM, dealers, direct) before building stock; teacher review of the syllabus mapping; pilot in two schools; price list; EMS RFQ.

9. Known limits

Simulated validation only; 10 of 20 experiments have a validated analysis; hardware unbuilt; sensor constants typical; USB bootloader flow untested.